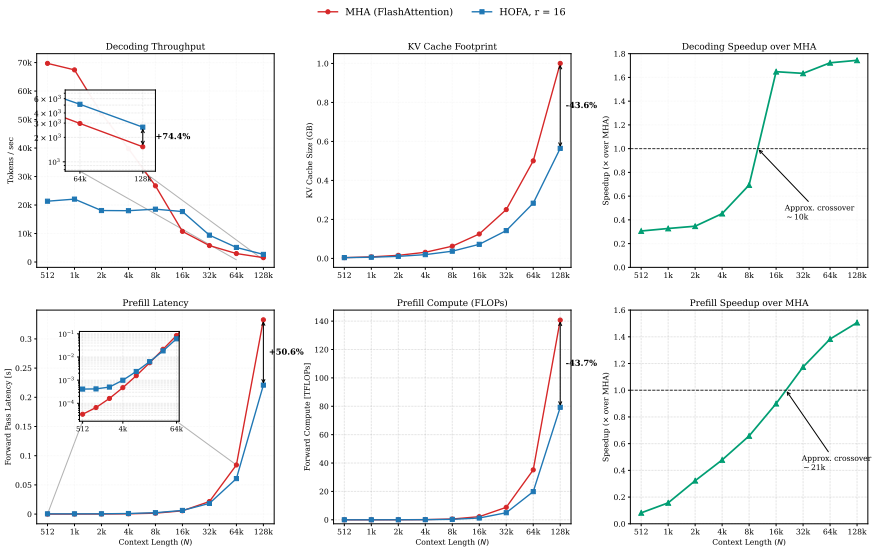
HOFA: Channel-Wise Hybrid Attention via Outlier Factorization

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Abstract

KV cache memory remains a primary bottleneck during long-context autoregressive decoding. While prior hybrid architectures mitigate this by alternating exact and linear attention across macro structural axes (depth, width, or sequence), they impose a coarse tradeoff between decoding efficiency and fine-grained associative recall. We introduce Hybrid Outlier-Factorized Attention (HOFA), a channel-wise intra-head decomposition. Inside every attention head, HOFA statically allocates a small r-dimensional outlier subspace to exact causal Softmax attention while routing the remaining inlier channels to a recurrent Gated Linear Attention (GLA) state. Single-layer hardware profiling demonstrates that this intra-head split reduces peak allocated KV-cache memory by 43.6% and yields up to a +74.4% decoding throughput speedup (1.74×) over standard MHA. Pretraining across 70M, 125M, and 350M scales shows that HOFA matches MHA perplexity, achieves competitive zero-shot reasoning, and, with learnable per-channel decay exponents, extrapolates to 64k context. We show that this factorization naturally concentrates global associative matching into the outlier channels while the linear recurrence absorbs background context, with retrieval limits closely following a proposed theoretical extreme-value capacity model.

https://github.com/LotanGutman/OutlierNeuronAttention



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Figure 1: **Hardware Execution & Throughput Benchmarks.** Empirical autoregressive decoding throughput and measured peak allocated KV-cache memory with compact $[B, H, N, r]$ key storage (top row) alongside prefill latency and attention-core FLOPs scaling (bottom row) comparing HOFA ($r = 16$) to FlashAttention-2 MHA.

1 Introduction & Related Work

Multi-Head Attention (MHA) [Vaswani et al., 2017] forms the computational core of modern Large Language Models (LLMs), but decoding is memory-bandwidth bottlenecked over long contexts. For a sequence length N, standard MHA must cache d_h-dimensional Key and Value vectors for every token across all heads and layers, incurring an $\mathcal{O}(NLHd_h)$ memory footprint per sequence. During generation ($N = 1$), memory bandwidth is dominated by streaming this large-scale KV-cache from High-Bandwidth Memory (HBM) into SRAM, limiting decoding throughput [Dao et al., 2023].

To address this memory bottleneck, recent research has explored hybrid architectures that combine exact $\mathcal{O}(N^2)$ Softmax attention with $\mathcal{O}(1)$ linear-time State-Space Models (SSMs) [Gu and Dao, 2023, Dao and Gu, 2024] or linear attention mechanisms [Peng et al., 2023, Yang et al., 2023, 2025]. Existing hybrid approaches decompose attention along three macro structural axes (Figure 2): (i) **depth-axis hybrids** alternating pure Softmax and linear layers sequentially across depth (e.g., StripedHyena [Poli et al., 2023], Jamba [AI21 Labs, 2024]); (ii) **width-axis hybrids** assigning distinct full-attention and linear-attention heads in parallel within each layer (e.g., Hymba [Dong et al., 2024]); and (iii) **sequence-axis sparsification** restricting exact Softmax to local sliding windows or selected token blocks (e.g., Longformer [Beltagy et al., 2020], Native Sparse Attention [Yuan et al., 2025]).

While existing hybrids multiplex across depth, width, or sequence dimensions (Figure 2a-c), HOFA introduces the fourth architectural axis: channel-wise intra-head decomposition (Figure 2d). Inside every attention head, HOFA statically splits the hidden dimension d_h into a small r-dimensional outlier channel $(\widetilde{Q}_{\mathcal{O}}, \widetilde{K}_{\mathcal{O}})$ and a remaining $(d_h - r)$-dimensional inlier channel $(\widetilde{Q}_{\mathcal{I}}, \widetilde{K}_{\mathcal{I}})$. The r outlier dimensions receive exact causal Softmax attention with Rotary Position Embeddings (RoPE) [Su et al., 2024], while the inlier dimensions are processed by a data-dependent GLA state. Unlike layer-wise hybrids such as Jamba [AI21 Labs, 2024] and Qwen3-Next [Qwen Team, 2025], which concentrate exact attention in selected layers and leave intervening layers purely recurrent, HOFA avoids an all-or-nothing layer-level decision by retaining a partial Softmax pathway at every layer. While sparse layer-wise schedules may be preferable when minimizing the number of full-attention layers is paramount, HOFA's design hypothesis is that maintaining reduced content-based retrieval at every layer benefits tasks requiring multi-depth associative recall.

This architectural factorization is motivated by systematic feature outliers in trained Transformers [Dettmers et al., 2022, Pierro and Abreu, 2024, An et al., 2025]. Quantization studies [Dettmers et al., 2022, Pierro and Abreu, 2024] demonstrate that extreme activation magnitudes cluster within a few systematic feature dimensions, which route specific semantic sub-tasks [An et al., 2025]. Optimization concentrates discriminative factual matching into a small subset of outlier channels $(\widetilde{Q}_{\mathcal{O}}, \widetilde{K}_{\mathcal{O}})$, while offloading background context to a recurrent GLA inlier state. Here, "outlier" refers to the high-magnitude feature subspace learned by the model; the allocation is architectural rather than determined by a runtime outlier detector. Prior KV-cache compression methods exploit channel structure post-hoc on pretrained dense models: LeanK [Zhang et al., 2025] learns static binary masks to drop key channels, ThinK [Xu et al., 2025] and SparK [Liao et al., 2026] dynamically select or prune-and-recover channels per query, and SparQ [Ribar et al., 2024] selects dominant query dimensions at each step to approximate scores for token fetching (complementing low-rank token selection in Loki [Singhania et al., 2024]). Rather than pruning channels post-hoc or discarding non-selected dimensions, HOFA enforces a static architectural channel partition during pretraining, jointly optimizing the exact Softmax subspace and the recurrent GLA inlier pathway.

We formalize our contributions across three core axes. First, we model HOFA's context capacity limits via the Welch bound, consistent with empirical induction sequence degradation. Second, hardware profiling on an NVIDIA RTX 5090 demonstrates a 43.6% peak KV-cache memory reduction and a 1.74× decoding throughput speedup over MHA. We motivate this architectural choice by analyzing the channel power distribution across production pretrained models. Finally, pretraining across 70M, 125M, and 350M scales demonstrates perplexity parity with MHA and strong zero-shot reasoning performance under extended 2× Chinchilla-optimal budgets [Hoffmann et al., 2022].

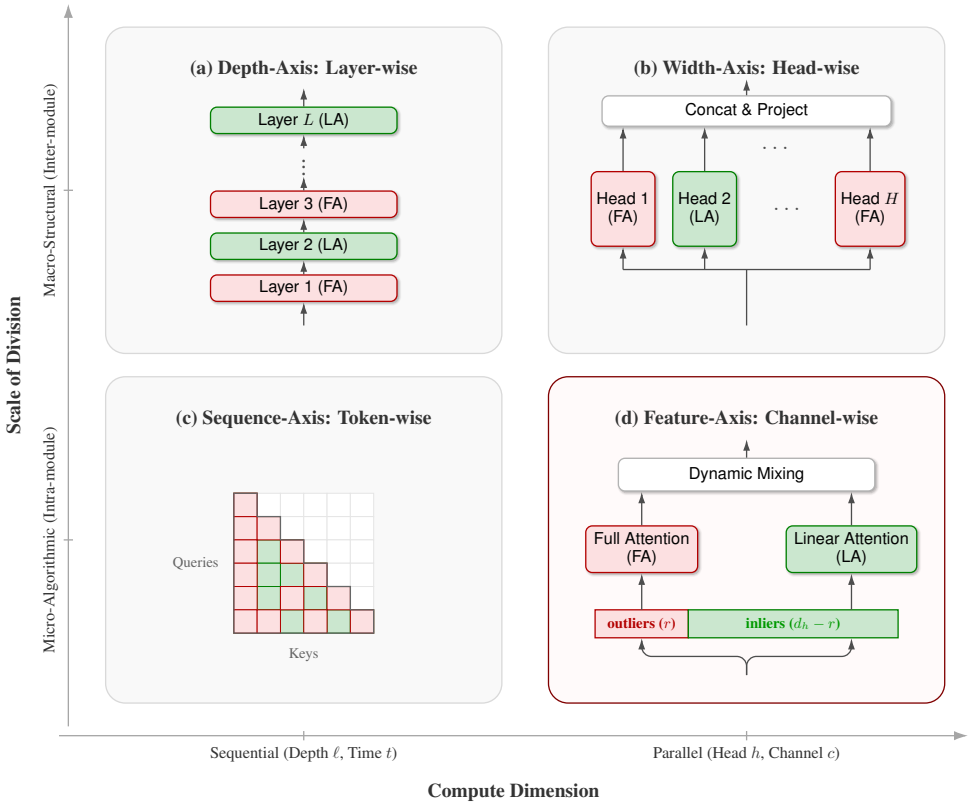


Figure 2: **The Architectural Taxonomy of Hybrid Attention.** Existing hybrid models decompose the attention operation along the (a) layer, (b) head, or (c) token dimension. HOFA introduces the (d) feature dimension axis, processing every token simultaneously through disjoint exact (**Softmax**) and linear (**GLA**) subspaces within the same head.

2 Methodology & Theoretical Framework

2.1 Architectural Formulation

Figure 3 illustrates the HOFA token mixer architecture. Inside every attention head, representations are factorized into isolated exact and linear processing streams. Following linear projection, Query and Key representations are statically partitioned into an r-dimensional outlier subspace $(\widetilde{Q}_{\mathcal{O}}, \widetilde{K}_{\mathcal{O}})$ and a $(d_h - r)$-dimensional inlier subspace $(\widetilde{Q}_{\mathcal{I}}, \widetilde{K}_{\mathcal{I}})$. The outlier stream executes exact causal Softmax attention with Rotary Position Embeddings (RoPE) to isolate high-magnitude associative matching, while the inlier stream drives a Gated Linear Attention (GLA) recurrence stabilized by RMSNorm [Zhang and Sennrich, 2019] and scaled by a learnable channel-wise scale vector $\mathbf{s}_{\mathcal{I}}$ to absorb diffuse background context. A learned, context-aware gate α_h dynamically blends both pathway outputs before final projection $\mathbf{W}_O$.

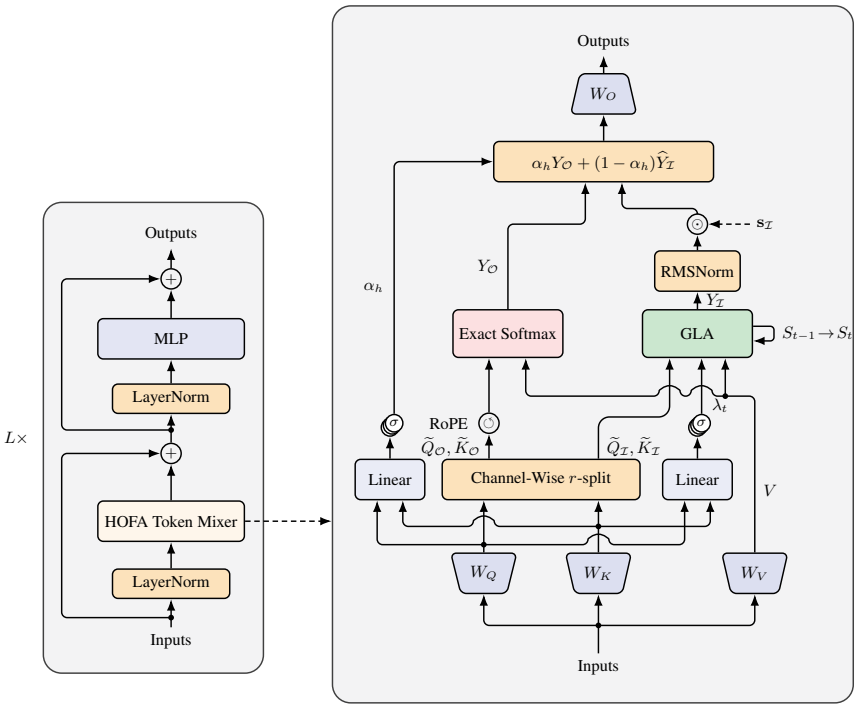
Let $\mathbf{X} \in \mathbb{R}^{B\times N\times d_{\text{model}}}$ denote the input sequence. For each attention head $h \in \{1, \dots, H\}$, query, key, and value representations are projected via linear weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_{\text{model}}\times d_h}$:

$$\widetilde{Q} = \mathbf{X}\mathbf{W}_Q, \quad \widetilde{K} = \mathbf{X}\mathbf{W}_K, \quad V = \mathbf{X}\mathbf{W}_V \quad \in \mathbb{R}^{B\times N\times d_h} \tag{1}$$

Inside each head, we partition the channels into the first r outlier dimensions and the remaining $j = d_h - r$ inlier dimensions:

$$\widetilde{Q} = \left[\widetilde{Q}_{\mathcal{O}} \parallel \widetilde{Q}_{\mathcal{I}}\right], \quad \widetilde{K} = \left[\widetilde{K}_{\mathcal{O}} \parallel \widetilde{K}_{\mathcal{I}}\right] \tag{2}$$

For the outlier pathway, we apply RoPE exclusively to the r-dimensional subspace: $\widehat{Q}_{\mathcal{O}} = \text{RoPE}(\widetilde{Q}_{\mathcal{O}})$, $\widehat{K}_{\mathcal{O}} = \text{RoPE}(\widetilde{K}_{\mathcal{O}})$. The outlier output $Y_{\mathcal{O}}$ is computed using exact causal Softmax



(a) Hybrid HOFA Transformer **(b) HOFA Token Mixer (intra-head)**

Figure 3: **Architecture of HOFA. (a)** Pre-LN Transformer layer containing the HOFA token mixer. **(b)** Inside the per-head mixer, projected queries and keys are statically split into r outlier channels (processed by exact causal Softmax attention) and remaining inlier channels (processed by Gated Linear Attention). Both pathways process the full value matrix V, with inlier outputs stabilized via RMSNorm and dynamically fused using a context-aware mixing gate α_h.

attention with variance normalization factor $S = \sqrt{d_h/r}$:

$$Y_\mathcal{O} = \text{softmax}\left(\frac{S}{\sqrt{d_h}}\widehat{Q}_\mathcal{O}\widehat{K}_\mathcal{O}^T + \mathbf{M}\right)V = \text{softmax}\left(\frac{1}{\sqrt{r}}\widehat{Q}_\mathcal{O}\widehat{K}_\mathcal{O}^T + \mathbf{M}\right)V \tag{3}$$

where $\mathbf{M}$ is the standard causal mask ($\mathbf{M}_{i,j} = 0$ for $i \geq j$, $-\infty$ otherwise).

For the inlier pathway, $\widetilde{Q}_\mathcal{I}$ and $\widetilde{K}_\mathcal{I}$ drive a GLA recurrence kernel with raw gate logits $a_t = \mathbf{W}_g[\widetilde{Q}_t, \widetilde{K}_t] + \mathbf{b}_g$ and retention gate $\lambda_t = \sigma(-a_t)$:

$$S_t = \lambda_t \odot S_{t-1} + \widetilde{K}_{\mathcal{I},t}^T V_t, \quad Y_{\mathcal{I},t} = \widetilde{Q}_{\mathcal{I},t} S_t \tag{4}$$

The inlier output is normalized via Root Mean Square Normalization (RMSNorm) and scaled by a learnable channel-wise inlier scale vector $\mathbf{s}_\mathcal{I} \in \mathbb{R}^{d_h}$ (initialized to 1):

$$\widehat{Y}_\mathcal{I} = \mathbf{s}_\mathcal{I} \odot \text{RMSNorm}(Y_\mathcal{I}) \tag{5}$$

Finally, both streams are dynamically blended via scalar gating $\alpha_h = \sigma(\mathbf{W}_\alpha[\widetilde{Q}, \widetilde{K}])$:

$$Y = \alpha_h \odot Y_\mathcal{O} + (1 - \alpha_h) \odot \widehat{Y}_\mathcal{I}, \quad \text{Output} = Y\mathbf{W}_O \tag{6}$$

2.2 Computational Complexity & Limiting Regimes

For sequence length N, layer count L, head count H, and hidden dimension d_h, standard causal MHA evaluates $QK^\top$ and AV over $\frac{1}{2}N^2$ causal token pairs, requiring $\text{FLOPs}_{\text{MHA,attn}}(N) = 2LHd_hN^2$ FLOPs. In HOFA, exact causal Softmax evaluates $Q_\mathcal{O}K_\mathcal{O}^\top$ on the r-dimensional outlier subspace, and multiplies attention weights by the full V, yielding an attention-core cost of $LH(r + d_h)N^2$.

Accounting for chunkwise inlier GLA recurrence ($\approx 6LH(d_h - r)d_h N$), total forward prefill FLOPs is:

$$\text{FLOPs}_{\text{HOFAPrefill}}(N) = 2PN + LH(r + d_h)N^2 + 6LH(d_h - r)d_h N \tag{7}$$

As $N \to \infty$, the quadratic term dominates, yielding the asymptotic attention-core prefill speedup ceiling relative to causal MHA:

$$\lim_{N\to\infty} \text{Speedup}_{\text{Prefill,attn}}(N) = \frac{2LHd_h N^2}{LH(r + d_h)N^2} = \frac{2d_h}{r + d_h} \tag{8}$$

For autoregressive decoding ($N = 1$), memory bandwidth becomes the primary bottleneck. HOFA mitigates this by storing only r-dimensional keys ($K_{\mathcal{O}}$) alongside full Values (V) and a compact $j \times d_h$ linear state (S), resulting in the footprint ratio:

$$\lim_{N\to\infty} \frac{\text{KV}_{\text{HOFA}}}{\text{KV}_{\text{MHA}}} = \frac{r + d_h}{2d_h} \tag{9}$$

where Equation 9 measures the per-token KV-cache ratio (excluding the fixed-size $j \times d_h$ recurrent state S), yielding the asymptotic decoding speedup limit:

$$\lim_{N\to\infty} \text{Speedup}_{\text{Decode}}(N) = \frac{2d_h}{r + d_h} \tag{10}$$

For $d_h = 128$ and $r = 16$, Equation 9 predicts a footprint ratio of $\frac{16+128}{256} = 0.5625$ (-43.8% memory reduction) and Equation 10 yields a maximum theoretical decoding throughput speedup ceiling of $\frac{256}{144} \approx 1.778\times$ (+77.8%).

Limiting Architectural Regimes. The HOFA formulation exhibits smooth mathematical continuity between two canonical limiting regimes. In **Pure Linear Attention** ($r = 0$), the outlier subspace vanishes, reducing output to the normalized inlier stream $Y = \mathbf{s}_{\mathcal{I}} \odot \text{RMSNorm}(Y_{\mathcal{I}})$; the quadratic vanishes, collapsing to exact linear attention compute of $6LHd_h^2 N$ FLOPs (or $(2P + 6LHd_h^2)N$ FLOPs for prefill). In **Dense Softmax Attention** ($r = d_h$), the inlier subspace vanishes, the linear term cancels to zero, and the quadratic term resolves to $LH(d_h + d_h)N^2 = 2LHd_h N^2$, recovering standard causal Multi-Head Attention FLOPs.

2.3 Capacity-Inspired Retrieval Model

We model HOFA's retrieval limits by framing the exact pathway as a continuous associative memory constrained by the aggregate interference floor of the Welch bound [Welch, 1974]. As derived in Appendix B, geometric cross-talk bounds the naive SNR ceiling by $\text{SNR}_{\text{naive}} \leq \frac{r}{cN-r}$, which isotropic scaling cannot overcome. Cross-entropy optimization drives an asymmetric gain factor γ within the r outlier channels against $n = N/d_h - 1$ competing distractors, yielding effective $\text{SNR}_{\text{eff}} \approx \gamma \frac{r}{cN-r}$ and logit mean $\mu = \sqrt{\text{SNR}_{\text{eff}}}$. Framing target retrieval as an M-ary hypothesis testing problem governed by Gumbel extreme value statistics [Fisher and Tippett, 1928] yields retrieval probability:

$$P(\text{correct}) \approx \Phi\left(\frac{\mu - b_n}{\sigma}\right) \tag{11}$$

where $b_n \approx \sqrt{2\ln(N/d_h - 1)}$ is the leading-order Gumbel threshold and σ is an empirical calibration parameter. Extended proofs and theoretical derivations are detailed in Appendix B.

3 Experimental Results

3.1 Hardware Execution & Efficiency

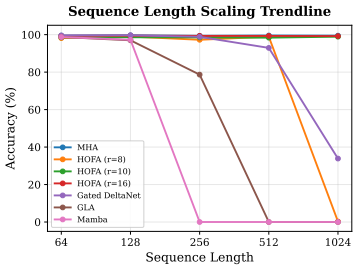
Experimental Settings. Appendix A (Table 2) details model configurations and full pretraining hyperparameters. All models are trained using PyTorch bfloat16 mixed-precision and AdamW (WD $= 0.1$) with Cosine Annealing. Depth-wise variation in pretrained channel power and effective dimension (d_{eff}) across model depth (Section 3.2) motivates investigating nonuniform capacity allocation across depth. Unless noted otherwise, we use a boundary-heavy U-shaped schedule

($r = 64, 32$ at boundary layers, tapering to $r = 16$ in intermediate layers) and compare it against uniform and increasing schedules under matched budgets in Section 3.3. Full layer-by-layer r-schedules for 70M ($L = 8$), 125M ($L = 12$), and 350M ($L = 24$) models are detailed in Appendix A (Table 3).

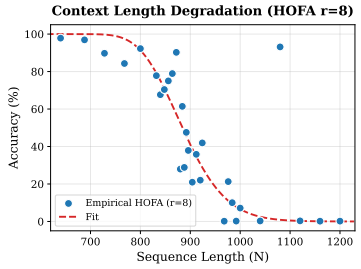
Computational Efficiency & Hardware Profiling. We benchmark HOFA against FlashAttention-2 MHA up to $N = 128$k on an NVIDIA RTX 5090 GPU using PyTorch mixed-precision (bfloat16) under dimensions $d_{\text{model}} = 2048, H = 16, d_h = 128, r = 16$. To isolate the core attention mechanism, we evaluate a single attention layer block with identity projections. During autoregressive decoding ($N = 1$), using compact $[B, H, N, r]$ key storage, HOFA reduces measured peak allocated KV-cache memory by 43.6% and sustains a +74.4% throughput speedup (1.74×) over MHA at 128k context length, crossing over at $N \approx 10$k (Figure 1, top), approaching the 1.78× theoretical bound (Equation 10). During prefill ($N > 1$), HOFA yields a ≈-43.8% attention-core forward FLOPs reduction and a +50.6% latency speedup at $N = 128$k tokens, crossing over at $N \approx 21$k, a gap that widens further as N scales toward the quadratic regime (Figure 1, bottom). Because this benchmark isolates a single attention block with identity projections, full-model end-to-end throughput also includes fixed projection, MLP, and vocabulary-head compute (with the heterogeneous schedules in Appendix A, Table 3 yielding theoretical KV-cache reductions of 39.6% at 125M and 41.1% at 350M).

3.2 Mechanistic Validation & Capacity Limits

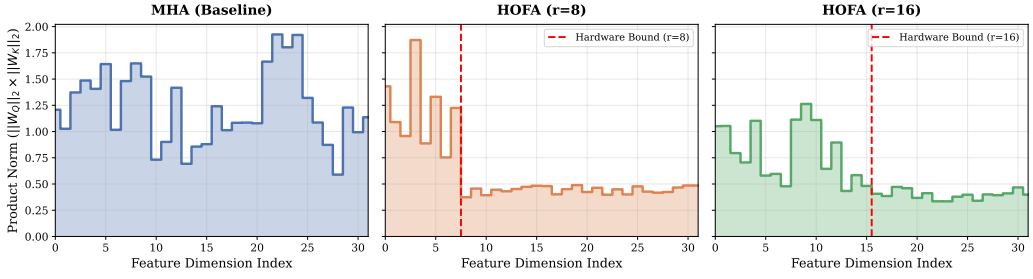
The Associative Recall Hierarchy. To evaluate sequence-scaling limits on associative recall, we benchmark 4-layer models ($L = 4, d_{\text{model}} = 128, H = 4, d_h = 32$) comparing Dense MHA, HOFA ($r = 8, 10, 16$), Gated DeltaNet, GLA, and Mamba-2 on synthetic Induction Head retrieval [Olsson et al., 2022] across context windows $N \in [640, 1200]$. As shown in Figure 4a, the evaluated architectures fall into four empirical performance regimes, which we refer to as an Associative Recall Hierarchy: Class I (Exact Softmax: MHA, HOFA $r \geq 10$) maintains full retrieval accuracy up to



(a) Sequence Length Memory Cliff



(b) Empirical vs. Calibrated Capacity Trendline ($r = 8$)



(c) Feature Norm Disparity Across Hidden Dimensions

Figure 4: **Mechanistic Validation & Capacity Dashboard.** (a) Associative recall accuracy vs. sequence length N across 4-layer models ($d_h = 32$). (b) Empirical degradation fitted with an empirically calibrated capacity trendline ($\hat{\gamma} = 16.11$, $\hat{\sigma} = 0.170$). (c) Parameter magnitude (norm) across hidden dimensions in trained weights, showing sharp outlier concentration at index $r - 1$ (dashed red lines mark static hardware boundaries $r = 8, 16$).

$N = 1024$; Class II (Dynamic Matrix Gated: Gated DeltaNet) delays memory decay via matrix updates, achieving 38% accuracy at $N = 1024$; Class III (Scalar Gated: GLA) transitions to 0% for $N \gtrsim 512$; and Class IV (Pure SSM: Mamba-2) reaches state capacity limits for $N \gtrsim 256$.

Empirical Capacity Bounds. Figure 4b evaluates length-specific $r = 8$ checkpoints trained and evaluated across 30 context lengths $N \in [640, 1200]$ via held-out inference with a fixed lookback gap, using average per-token accuracy as the empirical estimate of the single retrieval event probability modeled by Equation 11. Specifically, Figure 4b plots an empirically calibrated capacity trendline $P(\text{correct}) = 100 \cdot \Phi((\mu - b_n)/\sigma)$ with $n = cN - 1$, $b_n = \sqrt{2 \ln n}$, and $\mu = \sqrt{\gamma r/(cN - r)}$ using nonlinear least squares with fixed $c = 1/d_h = 1/32$, yielding $\hat{\gamma} = 16.11$ and $\hat{\sigma} = 0.170$. Accuracy shows a broad downward trend consistent with the theoretical extreme value capacity model, though individual checkpoints, including a localized peak near $N = 1080$, exhibit high variance consistent with the sampling sensitivity of synthetic associative recall tasks noted in Section 4.

Feature Norm Disparity. Probing parameter norms across trained induction task checkpoints provides empirical support for the asymmetric amplification mechanism (γ). Standard MHA scatters weight magnitude across all dimensions to capture RoPE frequencies, whereas HOFA ($r = 8, 16$) exhibits a sharp "norm cliff" precisely at index $r-1$, where we observe that optimization concentrates parameter magnitude into the r outlier channels (reaching ≈ 1.85 for $r = 8$) while leaving inlier channels at baseline levels (≈ 0.4, Figure 4c).

Pretrained Channel Power Distribution & Effective Dimension. To examine how query-key interaction energy distributes across feature coordinates in dense Transformers, we measure causal channel power E_c and normalized channel energy $p_c = E_c / \sum_{c'} E_{c'}$ across eight pretrained models from 124M to 8B parameters [Vig and Belinkov, 2019, Singhania et al., 2024, Xu et al., 2025] (Appendix C, Figure 7). Ranking channels by E_c shows that cumulative power $\mathcal{C}(r)$ concentrates rapidly within a small fraction of coordinates, while the Shannon effective dimension $d_{\text{eff}}(\ell) = \exp(-\sum_c p_c \ln p_c)$ captures how concentration varies across depth. Because sorted channel power in dense models is a post-hoc diagnostic rather than a direct measurement of HOFA's fixed-prefix architectural split [Zhang et al., 2025, Xu et al., 2025, Liao et al., 2026], we treat it as motivation for investigating depth-wise capacity allocation and evaluate both U-shaped and increasing capacity schedules under matched budgets (Appendix D).

3.3 End-to-End Language Modeling

Pretraining Dynamics. To ensure robust representation learning across the dual pathways, we pretrain 70M, 125M, and 350M models on FineWeb-Edu [Penedo et al., 2024] beyond compute-optimality at 40 tokens per parameter (2× Chinchilla [Hoffmann et al., 2022]). This corresponds to 2.8B, 5B, and 14B tokens at context length $N = 1024$. As shown in Figure 5, 350M HOFA closely tracks standard MHA throughout pretraining, matching final validation perplexity.

Dynamic Mixing Gate Analysis. To characterize the behavior of the dynamic mixing gate α_h, we extract layer-wise distributions from the 350M HOFA model across 1,000 FineWeb-Edu validation sequences (discarding the first 5 attention-sink tokens). Contrary to binary routing expectations,

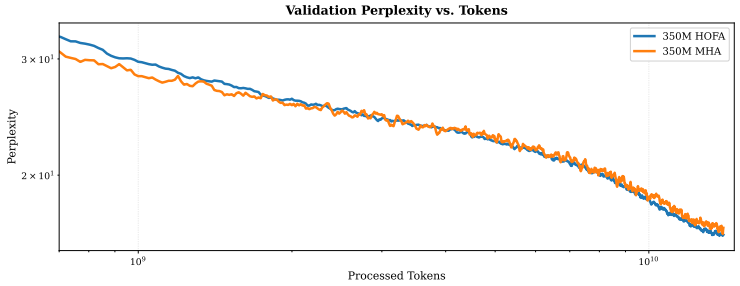


Figure 5: **350M Pretraining Dynamics.** Validation perplexity vs. processed tokens on FineWeb-Edu (14B tokens at $N = 1024$). 350M HOFA closely tracks standard MHA throughout training, matching final validation perplexity under identical token budgets.

Table 1: **Validation Perplexity and Zero-Shot Common-Sense Reasoning (%).** Evaluated across six common-sense reasoning benchmarks and on held-out FineWeb-Edu validation perplexity. All models, including external baselines, were re-evaluated locally under matched evaluation settings; external baselines report (—) for perplexity due to tokenizer vocabulary and pretraining domain incomparability. **Bold** denotes overall best in scale class; underline denotes best in 1:1 controlled MHA vs. HOFA pair. Higher is better (↑) for reasoning tasks; lower is better (↓) for perplexity.

Model	Tokens	Val PPL ↓	ARC-E ↑	ARC-C ↑	PIQA ↑	Wino ↑	OBQA ↑	Hella ↑
125M MHA	5B	23.25	**46.2**	24.8	61.3	49.7	29.4	31.2
125M HOFA	5B	22.99	44.9	**26.5**	61.2	51.7	**30.2**	31.8
GPT-2 (124M)	10B	—	39.5	22.7	62.5	51.6	27.2	31.1
Pythia-160M	300B	—	39.4	23.6	61.2	52.1	26.8	30.1
Mamba-130M	300B	—	41.9	24.3	**63.2**	**52.5**	28.6	**35.2**
350M MHA	14B	15.88	50.4	26.3	66.5	51.1	32.2	41.0
350M HOFA	14B	15.87	**53.8**	**28.8**	66.8	51.5	**34.2**	42.1
OPT-350M	180B	—	40.1	23.7	64.4	52.4	28.4	36.8
Pythia-410M	300B	—	46.0	24.4	67.1	53.5	29.4	40.6
Mamba-370M	300B	—	48.2	27.9	**68.4**	**55.6**	31.2	**46.5**

token-level gate values span the full $[0, 1]$ range dynamically across sequence positions, while layer-wise mean gate values across middle and deep layers cluster tightly in a soft-blending regime ($\mu_\alpha \approx 0.38$–0.55, averaging ~ 0.45). Layer 1 exhibits a unique, low-variance concentration at $\mu_\alpha \approx 0.24$, heavily prioritizing the recurrent GLA pathway to establish initial local context (see Appendix F, Figure 10).

Effective Attention Distance: Preservation of Global Reach. To test whether confining exact Softmax to r dimensions limits global attention reach, we measure layer-wise effective attention distance across 1,000 FineWeb-Edu validation sequences. Appendix G (Figure 11) shows 350M HOFA's outlier pathway tracking standard MHA ($d_h = 128$) across all 24 layers (averaging 257.1 vs. 255.9 tokens, near the $N/4 = 256$ uniform-causal baseline at $N = 1024$). Conversely, the GLA inlier pathway operates at a sharp local average of 3.9 tokens under a magnitude-normalized pseudo-probability metric (Appendix G). This indicates that HOFA preserves MHA's layer-wise reach profile in the outlier subspace while the recurrent linear state primarily handles immediate local context.

Language Modeling and Zero-Shot Evaluation. We benchmark HOFA against standard MHA by measuring validation perplexity, alongside zero-shot performance against established baselines across six common-sense tasks using `lm-evaluation-harness` [Gao et al., 2021]. As detailed in Table 1, 125M HOFA outperforms matched 125M MHA across 4 of 6 benchmarks while remaining competitive with academic baselines such as GPT-2 [Radford et al., 2019], Pythia-160M [Biderman et al., 2023], and Mamba-130M [Gu and Dao, 2023]. At the 350M parameter scale, HOFA outperforms matched 350M MHA across all six benchmarks, while scoring higher on ARC-E, ARC-C, and OBQA than higher-token baselines like OPT-350M [Zhang et al., 2022], Pythia-410M [Biderman et al., 2023], and Mamba-370M [Gu and Dao, 2023].

Long Context Extrapolation and Channel Wise Decay ($\boldsymbol{\beta}$). To evaluate context length extension beyond the $N = 1024$ pretraining window, we measure perplexity on the 54 PG-19 test books with at least 64k tokens [Rae et al., 2020] from 1k to 64k using YaRN RoPE scaling [Peng et al., 2024] ($s = 16$) on our 125M models, both without adaptation (base) and after continual pretraining (CPT) at $N = 16384$. In HOFA, YaRN rescales rotary frequencies exclusively within the r-dimensional outlier subspace, leaving the recurrent inlier channels governed by their decay gate. Without adaptation (Figure 6, left), base HOFA outperforms base MHA across extended lengths. Following CPT at $N = 16384$ (Figure 6, right), scalar HOFA trails MHA; we hypothesize this occurs because a single scalar retention gate λ_t forces all j inlier channels in a head to share one decay rate, slowing adaptation to longer contexts. Introducing a learnable exponent $\boldsymbol{\beta}_h \in \mathbb{R}^j_{>0}$ per head (parameterized in log space, initialized to $\mathbf{1}$, yielding $\lambda_{t,c} = \lambda_t^{\beta_{h,c}}$) allows HOFA ($\boldsymbol{\beta}$) to adapt channel-specific forgetting rates, outperforming both MHA and scalar HOFA across every context length from 1k to 64k while maintaining stability across pre-softmax temperature scales (Appendix E).

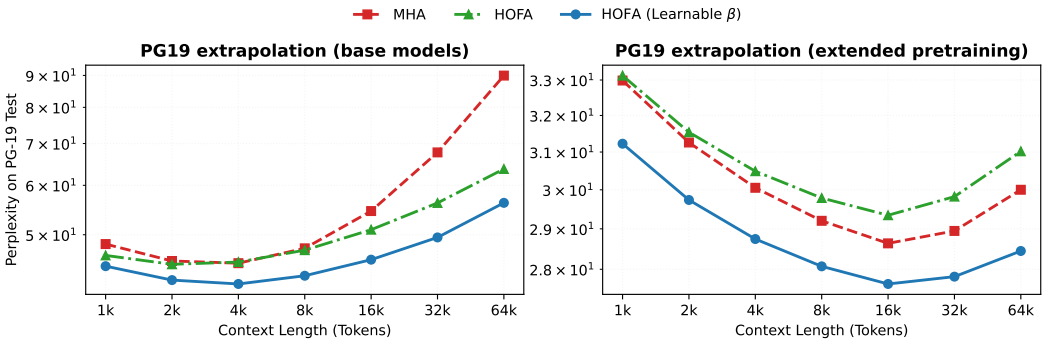


Figure 6: **125M PG-19 Long Context Extrapolation.** Perplexity from 1k to 64k for base models without adaptation (left) and after $N = 16384$ YaRN CPT (right). HOFA (Channel Wise $\boldsymbol{\beta}$) achieves the lowest perplexity across all evaluated lengths in both regimes.

Macro-Axis Structural Controls. Pretraining a 70M parameter ablation suite ($2\times$ Chinchilla-optimal budget) under matched parameter counts (with all hybrid and control variants sharing a matched 256-channel exact-attention key budget) shows that intra-head HOFA achieves lower validation perplexity than Depth-Axis (Jamba-style) and Width-Axis (Hymba-style) controls and edges them out on average zero-shot accuracy (single-seed; Boundary-Matched Depth-Axis leads on ARC-E and WinoGrande), with context-aware gating contributing positively to pathway fusion. Specifically, U-shaped HOFA (30.65 Val PPL, 39.15% zero-shot avg) outperforms Width-Axis (31.31 PPL, 38.58%), Interior Depth-Axis (31.54 PPL, 38.03%), and Boundary-Matched Depth-Axis (30.86 PPL, 38.92%) controls, while Increasing HOFA achieves the lowest validation perplexity (30.35 PPL, 38.85% avg). Full ablation curves and configurations are detailed in Appendix D.

4 Conclusion & Limitations

Limitations & Future Work. While HOFA establishes an effective channel-wise hybridization scheme, several limitations should be noted. First, native pretraining is evaluated up to the 350M parameter scale on a single-node setup; scaling to billion-parameter regimes remains for future work. Second, static r_ℓ-schedules are guided by post-hoc channel power diagnostics (Section 3.2), leaving adaptive or learnable per-layer r_ℓ allocation for future study. Third, pretraining runs reflect single-seed executions due to compute limits. Finally, synthetic associative recall tasks can be prone to sampling sensitivity; we position block copying benchmarks (Appendix H) strictly as mechanistic probes, grounding our core claims in pretraining dynamics and zero-shot reasoning. Looking ahead, extending channel hybridization to composite compression schemes (Appendix I) presents promising directions for extreme long-context scaling, while deep, narrow-head HOFA architectures offer attractive design points for edge inference [Dao et al., 2023, Jiang et al., 2025].

Conclusion. We introduce HOFA, an intra-head hybrid attention layer that factorizes the hidden dimension into exact and recurrent processing streams. A dedicated r-dimensional Softmax subspace isolates and retrieves high-magnitude semantic features, while a Gated Linear Attention state processes the remaining channels to absorb background context. This decouples global associative recall from temporal state decay, recovers both pure linear and dense attention as limiting architectural cases, and enables the exact-attention pathway to preserve long-range associative retrieval while the linear pathway handles local and background context. Under matched pretraining up to 350M parameters, HOFA matches the dense MHA baseline across language modeling, commonsense reasoning, and synthetic retrieval, while reducing the decoding KV-cache footprint and improving peak throughput. Mechanistic analyses show that both pathways contribute to the hybrid state, with optimization naturally concentrating the long-range associative capacity into the Softmax outlier channels.

Reproducibility Statement. Hyperparameters and theoretical proofs are documented in Appendices A and B. Custom Triton kernels are detailed in Appendix J, with full pretraining and evaluation code available at `https://github.com/LotanGutman/OutlierNeuronAttention`.

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A Full Hyperparameter & Model Configuration Tables

All profiling and pretraining benchmarks were executed on an NVIDIA RTX 5090 GPU (requiring approximately 4 days for the 350M parameter pretraining run). Table 2 lists full pretraining hyperparameters, and Table 3 details the exact layer-by-layer heterogeneous r-schedules across model scales.

Table 2: **Full Pretraining Hyperparameters.**

Parameter	**70M**	**125M**	**350M**
Layers (L)	8	12	24
Hidden Dim (d_{model})	512	768	1024
Heads (H)	8	6	8
Training Tokens	2.8B	5.0B	14.0B
Batch Size (sequences)	32	32	32
Sequence Length (N)	1024	1024	1024
Learning Rate	6e-4	5e-4	3e-4
Optimizer	AdamW	AdamW	AdamW
Weight Decay	0.1	0.1	0.1
Adam β_1, β_2	0.9, 0.95	0.9, 0.95	0.9, 0.95
Warmup Steps	500	750	2000
Precision	bfloat16	bfloat16	bfloat16

Table 3: **Heterogeneous Outlier Routing Schedules (r-schedules) Across Model Scales.**

Model Scale	**Layers (L)**	**Layer-by-Layer Outlier Dimension Schedule (r)**
70M	8	$[64, 32, 16, 16, 16, 16, 32, 64]$
125M	12	$[64, 32, 16, 16, 16, 16, 16, 16, 16, 16, 32, 64]$
350M	24	$[64, 32, 32, \underbrace{16, \ldots, 16}_{\times 18}, 32, 32, 64]$

B Extended Mathematical Proofs & Derivations

The Welch Bound and the Scaling Paradox. Packing N key vectors into a restricted r-dimensional routing subspace creates inevitable geometric interference. By Welch's theorem [Welch, 1974], the aggregate off-diagonal squared inner products of N unit vectors satisfy:

$$\sum_{i=1}^{N} \sum_{j \neq i}^{N} |\langle x_i, x_j \rangle|^2 \geq \frac{N^2}{r} - N = \frac{N(N-r)}{r} \tag{12}$$

Whereas classical coding theory often considers pairwise maximum correlation [Welch, 1974], associative retrieval in attention aggregates interference across all $N-1$ competing tokens simultaneously. The expected per-vector interference power seen by a query vector is:

$$I = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq i}^{N} |\langle x_i, x_j \rangle|^2 \geq \frac{N-r}{r} \tag{13}$$

which scales linearly with sequence length N. Assuming unit signal alignment for the target key ($\langle q, k_0 \rangle^2 = 1$), Equation 13 dictates that the naive Signal-to-Noise Ratio is bounded from above by an achievable capacity ceiling:

$$\text{SNR}_{\text{naive}} = \frac{1}{I} \leq \frac{r}{N-r} \tag{14}$$

To model structured representations in multi-head attention, we introduce the semantic subspace partitioning hypothesis: keys are distributed across d_h approximately orthogonal feature directions,

concentrating query competition into an active subspace pool of cN tokens with distractor density $c = 1/d_h$. Within this shared semantic subspace, the naive SNR ceiling becomes:

$$\text{SNR}_{\text{naive}} \leq \frac{r}{cN - r} \tag{15}$$

While continuous interference variance scales with $(cN - r)$, discrete target retrieval faces $n = cN - 1 = N/d_h - 1$ competing distractors. This establishes an apparent scaling paradox: if the network attempts to boost signal separation by uniformly scaling projection weights by an isotropic magnitude factor S ($q \leftarrow S\hat{q}, k \leftarrow S\hat{k}$), target signal power scales as S^2, but distractor noise variance scales proportionally as $S^2 I \geq S^2 \frac{cN-r}{r}$. The factor S^2 cancels algebraically in the SNR ratio, leaving the naive SNR ceiling strictly invariant to uniform magnitude scaling.

Asymmetric Gradient Gain. We model HOFA's feature crystallization as an asymmetric optimization process. Cross-entropy loss rewards increasing the true target logit z_0 relative to distractors z_i, applying an asymmetric, anisotropic gradient pressure. The optimizer selectively amplifies discriminative subspace directions responsible for pattern matching, while ambient distractor directions receive diffuse gradients under the Softmax partition function. We define $\gamma \gg 1$ as this loss-driven asymmetric gain factor, yielding the effective Signal-to-Noise Ratio:

$$\text{SNR}_{\text{eff}} \approx \gamma \frac{r}{cN - r}, \quad \mu = \sqrt{\text{SNR}_{\text{eff}}} \approx \sqrt{\gamma \frac{r}{cN - r}} \tag{16}$$

This anisotropic gain is empirically manifested by the sharp "norm cliff" in outlier feature projections (Section 3.2, Figure 4c), where optimization concentrates high-magnitude associative matching into the r outlier channels.

M-ary Hypothesis Testing & The Gumbel Threshold. For successful retrieval, the target logit $z_0 \sim \mathcal{N}(\mu, 1)$ must exceed the maximum distractor logit $M_n = \max_{i=1}^{n} z_i$ across $n = N/d_h - 1$ competing keys. Under the Fisher-Tippett-Gnedenko theorem [Fisher and Tippett, 1928], the maximum of n Gaussian distractors converges to a Gumbel distribution with leading-order threshold $b_n \approx \sqrt{2 \ln n}$. The retrieval probability is:

$$P(\text{correct}) = P(z_0 > M_n) \approx \Phi\left(\frac{\mu - b_n}{\sigma}\right) \tag{17}$$

Here, σ is an empirical calibration parameter. While standard extreme value theory assumes independent unit-variance distractors, real transformer representations exhibit feature correlations and non-unit variance; σ calibrates the effective transition width without altering the underlying Gumbel scaling trajectory.

C Channel Power Distribution and Effective Dimension Across Production Models

To quantify how causal query-key channel energy concentrates across head dimensions in pretrained Transformers, we measure the per-channel QK-power proxy per feature dimension $c \in \{1, \ldots, d_h\}$ for each head:

$$E_c = \frac{1}{d_h} \sum_{i=1}^{N} Q_{i,c}^2 \left(\sum_{j \leq i} K_{j,c}^2 \right) \tag{18}$$

Normalizing across feature channels yields a scale invariant probability distribution p_c:

$$p_c = \frac{E_c}{\sum_{c'=1}^{d_h} E_{c'}} \tag{19}$$

From p_c, we compute two quantities:

- **Cumulative Channel Power**: The cumulative distribution of channels sorted in descending order of probability ($p^{(1)} \geq p^{(2)} \geq \cdots \geq p^{(d_h)}$),

$$\mathcal{C}(r) = \sum_{k=1}^{r} p^{(k)} \in [0, 1]. \tag{20}$$

- **Effective Dimension**: The Shannon perplexity of p_c at layer ℓ,

$$d_{\text{eff}}(\ell) = \exp\left(-\sum_{c=1}^{d_h} p_c \ln(p_c + 10^{-12})\right), \tag{21}$$

which is bounded in $[1, d_h]$: it attains $d_{\text{eff}} = 1$ when all energy concentrates in a single channel, and $d_{\text{eff}} = d_h$ when energy is distributed uniformly across all channels.

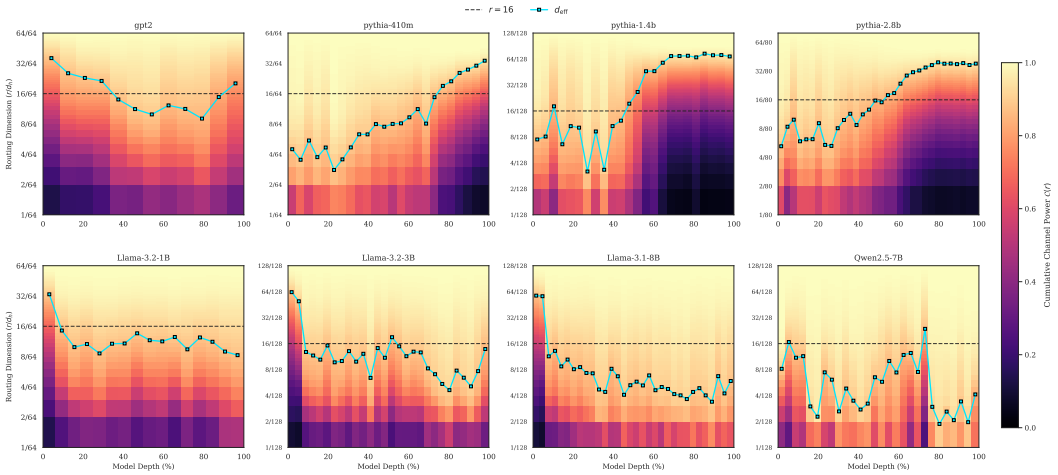


Figure 7: **Channel Power Concentration and Effective Dimension Across 8 Production Models.** Cumulative sorted channel power $\mathcal{C}(r)$ and layer-wise effective dimension $d_{\text{eff}}(\ell)$ across pure MHA models (GPT-2 124M, Pythia 410M/1.4B/2.8B) and GQA models (Llama-3.2 1B/3B, Llama-3.1-8B, Qwen2.5-7B), with the $r = 16$ reference line.

D Extended Ablation Suite & Macro-Axis Controls

To evaluate the intra-head channel-wise hybridization axis against macro-structural alternatives, we pretrain a 70M parameter ablation suite ($L = 8, d_{\text{model}} = 512, H = 8, d_h = 64$) on 2.8B tokens ($2\times$ Chinchilla-optimal budget, 40 tokens/parameter). Zero-shot reasoning benchmarks (Table 4) and validation perplexity curves (Figure 8) reflect matched parameter counts across all 70M variants, with all hybrid and control variants sharing a matched 256-channel exact-attention key budget.

Capacity Matching (Average $r = 32$). To establish a fair comparison, all macro axis baselines are parameterized to match Canonical HOFA's total exact attention capacity. Canonical HOFA employs

Table 4: **70M Parameter Scale Validation Perplexity and Zero-Shot Common-Sense Reasoning (%).** Evaluated across six common-sense reasoning benchmarks and on held-out FineWeb-Edu validation perplexity. All models are pretrained under matched parameter counts ($2\times$ Chinchilla-optimal, 2.8B tokens); all hybrid and control variants share a matched 256-channel exact-attention key budget (50% of the 512 exact channels in the full MHA baseline). **Bold** denotes best performance. Higher is better (↑) for reasoning tasks; lower is better (↓) for perplexity.

Model (70M Scale)	Val PPL ↓	ARC-E ↑	ARC-C ↑	PIQA ↑	Wino ↑	OBQA ↑	Hella ↑
MHA Baseline	31.37	41.2	**25.4**	57.7	50.6	28.6	28.2
Reduced-QK MHA ($d_{QK} = 32$)	30.99	**41.7**	25.1	58.4	47.9	28.2	28.3
Depth-Axis Hybrid (Interior)	31.54	40.8	22.7	59.1	48.8	28.8	28.0
Depth-Axis Hybrid (Boundary)	30.86	41.4	23.8	58.0	**52.1**	30.0	28.2
Width-Axis Hybrid (GLA + MHA)	31.31	40.4	23.3	59.0	50.9	30.2	27.7
HOFA (Flat $r = 32$ + Dynamic Gate)	30.96	41.1	24.1	57.9	50.8	30.6	27.8
Canonical HOFA (U-Shaped + Dynamic Gate)	30.65	40.6	23.7	**60.2**	51.1	**30.8**	**28.5**
HOFA (Increasing r + Dynamic Gate)	**30.35**	41.0	23.7	59.9	51.1	29.4	28.0
Static-Blend HOFA (− Dynamic Gate)	30.85	39.7	24.6	58.8	51.1	30.6	28.2

a heterogeneous, boundary heavy routing schedule ($[64, 32, 16, 16, 16, 16, 32, 64]$; Table 3), yielding a cumulative budget of 256 exact attention channels across 8 layers (an average of $r = 32$ per layer). Consequently, the Uniform HOFA control fixes $r = 32$ across all layers, while the Depth-Axis and Width-Axis hybrids allocate exactly 50% of their total capacity (256 out of 512 total channels) to exact MHA, ensuring matched exact-key channel counts (256 exact key channels across 8 layers) and closely matched checkpoint architectures (76,687,488 stored parameters for dynamic-gate HOFA, MHA, Reduced-QK MHA, and Depth-Axis; 76,679,296 for Static-Blend HOFA; and 76,679,232 for Width-Axis Hybrid; in the forward pass, Reduced-QK MHA slices W_Q, W_K to $d_{QK} = 32$ for ≈ 74.6M active parameters, and MHA/Depth-Axis omit the ≈ 0.02M gating projections). Note that because HOFA's exact branch processes full d_h-dimensional values, its quadratic attention-core term scales as $(r + d_h)$ rather than d_h in a 50% MHA hybrid ($1.5\times$ at $r = d_h/2$), though dense MLP projections dominate total pretraining compute at $N = 1024$.

Controlled Depth-Axis Baselines. To separate hybridization axis from layer placement, we evaluate two 50%-capacity Depth-Axis (Jamba-style) controls alternating four pure MHA layers ($d_h = 64$) and four pure GLA layers ($d_h = 64$): an interior schedule ($[0, 64, 64, 0, 0, 64, 64, 0]$) and a boundary-matched schedule ($[64, 0, 64, 0, 0, 64, 0, 64]$) that places exact attention at Layers 0 and 7. Placing full attention at the boundaries improves the depth hybrid from 31.54 to 30.86 validation perplexity ($38.03\% \rightarrow 38.92\%$ zero-shot average), confirming the value of boundary exact capacity; however, intra-head HOFA still achieves lower validation perplexity (30.65 U-shaped, 30.35 increasing) and higher overall zero-shot reasoning accuracy (39.15%), and these single-seed results are consistent with a benefit from distributing exact-attention capacity within each layer.

Micro-Architecture Control (GLA vs. Mamba). Across all hybrid baselines (Depth-Axis and Width-Axis), the linear pathway is implemented using our exact Gated Linear Attention (GLA) kernel. Holding the linear micro-architecture constant isolates the structural hybridization axis (layer-wise vs. head-wise vs. channel-wise) as the sole independent variable.

Static vs. Dynamic Blending. In addition to macro axis baselines, we evaluate a Static-Blend HOFA variant where the dynamic token-level mixing gate α_h is replaced by a fixed, layer-wise scalar, isolating the contribution of context-aware dynamic gating.

Reduced-QK MHA Control. To test whether HOFA's recurrent inlier stream contributes beyond key-dimension truncation [Singhania et al., 2024, Xu et al., 2025], we train a low-rank query/key MHA baseline ($d_{QK} = 32$ channels, full $d_h = 64$ values, zero inlier recurrence) at an identical 96-element per-head KV-cache footprint. Reduced-QK MHA achieves 30.99 validation perplexity (38.27% zero-shot average), trailing both Flat HOFA (30.96 PPL, 38.72% avg) and Canonical U-shaped HOFA (30.65 PPL, 39.15% avg), with notable drops on WinoGrande (47.9% vs. 51.1%) and OpenBookQA (28.2% vs. 30.8%). Under these matched conditions, HOFA outperforms pure query/key truncation.

Increasing vs. U-Shaped Schedules. To test an alternative depth allocation under the identical 256-channel cumulative budget ($\sum r = 256$), we evaluate an Increasing Schedule ($r = [16, 16, 16, 16, 32, 48, 48, 64]$), shifting 64 channels from Layers 0–1 to Layers 4–6 [Vig and Belinkov, 2019, Cai et al., 2025]. Empirically, the Increasing Schedule achieves the lowest validation perplexity across all variants (30.35 vs. 30.65 for Canonical U-shaped and 31.37 for MHA), whereas Canonical U-shaped HOFA retains a higher overall zero-shot reasoning average (39.15% vs. 38.85%), including OpenBookQA (30.8% vs. 29.4%) and HellaSwag (28.5% vs. 28.0%).

Ablation Results & Analysis. Results from our ablation suite, detailed in Table 4 and Figure 8, demonstrate the advantages of channel-wise hybridization. While the Homogeneous ($r = 32$) HOFA baseline performs competitively, upgrading to the heterogeneous, boundary-heavy U-shaped r-schedule yields higher downstream accuracy. Under matched cumulative budgets, allocating higher exact capacity to boundary layers improves zero-shot reasoning over a flat distribution, whereas shifting capacity toward deeper layers (Increasing Schedule) further lowers validation perplexity. Canonical HOFA achieves lower validation perplexity (30.65 vs. 30.86–31.54) and a higher average zero-shot reasoning accuracy (39.15% vs. 38.03%–38.92%) than all Depth-Axis and Width-Axis hybrids, while optimizing faster to reach target perplexities earlier under matched token budgets (Figure 8). Finally, removing the dynamic mixing gate (Static-Blend HOFA) degrades accuracy,

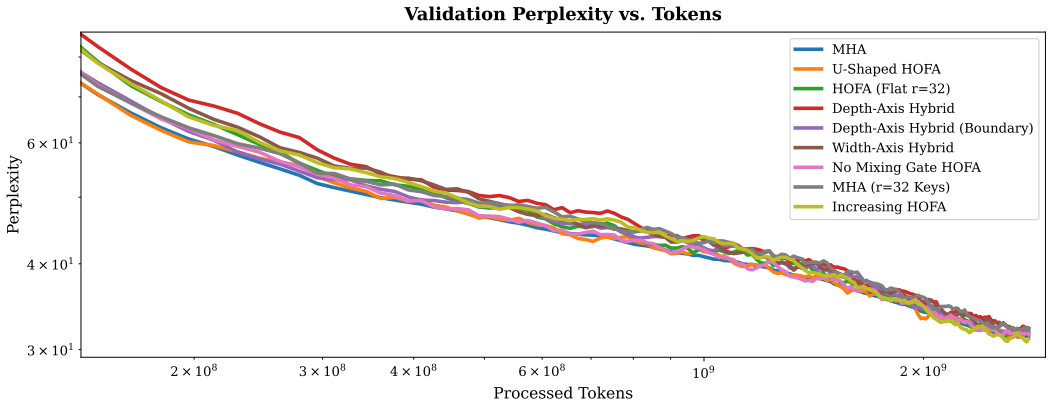


Figure 8: **Validation Perplexity Curves for the 70M Parameter Ablation Suite.** U-Shaped HOFA and Static-Blend HOFA track the pure MHA baseline, whereas the Depth-Axis and Width-Axis hybrids exhibit slower initial optimization during the first 10^9 tokens (rolling smoothing applied).

indicating that dynamic per-token routing projection benefits exact-linear pathway fusion over a static scalar. We note that these controls isolate the hybridization axis at 70M scale under matched linear-attention operators and 50% key budgets; benchmarking against sparser layer-wise schedules is left to larger-scale evaluation, so the advantage of per-depth exact routing in that regime remains a design hypothesis.

E Long Context Extrapolation and Channel Wise Decay ($\boldsymbol{\beta}$)

To complement the PG-19 extrapolation results in Section 3.3 (Figure 6), we evaluate sensitivity to pre-softmax attention temperature scaling $1/\sqrt{t} \in [1.0, 1.5]$ across 2k segments at $N = 16384$ after YaRN CPT ($s = 16$) for MHA, Scalar Gate HOFA (λ_t), and HOFA with Channel Wise decay exponents $\boldsymbol{\beta}_h \in \mathbb{R}^j_{>0}$ per head (Figure 9). Both scalar HOFA and HOFA ($\boldsymbol{\beta}$) exhibit nearly identical stability across $1/\sqrt{t} \in [1.0, 1.5]$, avoiding the sharp perplexity spike of MHA at $1/\sqrt{t} = 1.0$ because only the r outlier channels undergo Softmax temperature scaling while the j recurrent inlier channels remain invariant.

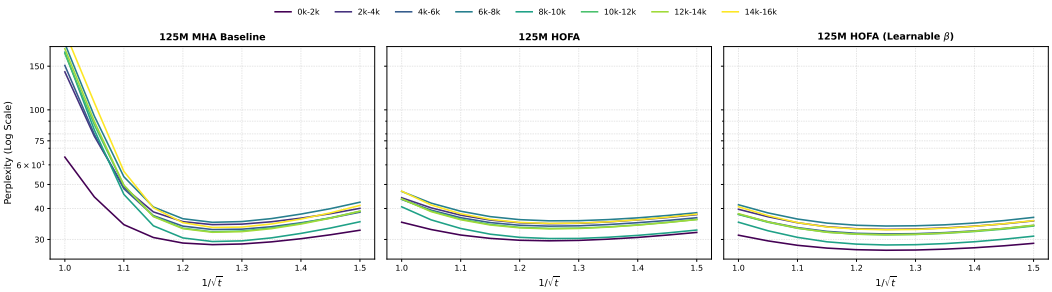


Figure 9: **Pre-Softmax Temperature Scaling Sweep ($1/\sqrt{t} \in [1.0, 1.5]$).** Perplexity across 2k segments at $N = 16384$ on PG-19 for MHA, Scalar HOFA, and HOFA (Channel Wise $\boldsymbol{\beta}$). Both HOFA variants remain flat and stable across the temperature sweep, avoiding MHA's perplexity spike at $1/\sqrt{t} = 1.0$.

F Dynamic Mixing Gate Distribution

To characterize dynamic mixing gate behavior, we extract layer-wise distributions from the 350M HOFA model ($L = 24$) across 1,000 validation sequences. As shown in Figure 10, token-level gate values span the full $[0, 1]$ range, demonstrating dynamic, context-dependent blending rather than hard

binary routing. Across middle and deep layers, mean gate values cluster in a soft-blending regime. Layer 1 displays an anomalous low-variance distribution, reflecting a strong initial reliance on the recurrent GLA state for local context aggregation.

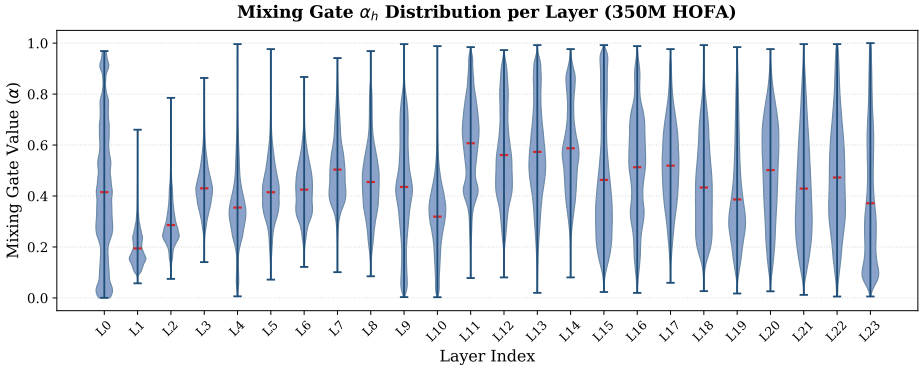


Figure 10: **Mixing Gate α_h Distribution per Layer (350M HOFA).** Layer-wise empirical distribution of the dynamic mixing gate across 24 layers. Token-level gate values span $[0, 1]$ dynamically, while layer-wise means across middle and deep layers cluster in a soft-blending regime ($\mu_\alpha \approx 0.38$–0.55, horizontal ticks). Layer 1 exhibits anomalous, low-variance reliance on the inlier GLA pathway ($\mu_\alpha \approx 0.24$).

G Effective Attention Distance Analysis

To evaluate whether HOFA maintains long range context dependency comparable to full Softmax attention, we compute the layer wise effective attention distance d_l across 1,000 validation sequences from FineWeb-Edu following standard attention distance methodologies. For standard MHA and HOFA's exact outlier pathway, d_l is defined as:

$$d_l^{\text{outlier}} = \frac{1}{N-5} \sum_{i=6}^{N} \sum_{j \le i} p_{ij}^{(l)} (i-j), \quad \text{where } p_{ij}^{(l)} = \text{softmax}_{j \le i} \left(\frac{1}{\sqrt{r}} \widetilde{Q}_{\mathcal{O},i}^{(l)} (\widetilde{K}_{\mathcal{O},j}^{(l)})^T + M_{ij} \right) \quad (22)$$

discarding the first 5 attention-sink tokens [Xiao et al., 2023]. Because the unrolled linear attention state yields unbounded and potentially negative values, we introduce a magnitude-normalized pseudo-probability $P_{ij}^{(l)}$ to enable direct distance comparisons with Softmax distributions. This is derived from the inlier feature content $A_{ij}^{(l)}$ and the cumulative gate decay matrix $D_{ij}^{(l)}$:

$$A_{ij}^{(l)} = \widetilde{Q}_{\mathcal{I},i}^{(l)} (\widetilde{K}_{\mathcal{I},j}^{(l)})^T, \qquad D_{ij}^{(l)} = \prod_{s=j+1}^{i-1} \lambda_s^{(l)} = \prod_{s=j+1}^{i-1} \sigma(-a_s^{(l)}) \quad (j < i), \quad (23)$$

$$P_{ij}^{(l)} = \frac{|D_{ij}^{(l)} A_{ij}^{(l)}|}{\sum_{k<i} |D_{ik}^{(l)} A_{ik}^{(l)}|}, \qquad d_l^{\text{inlier}} = \frac{1}{N-5} \sum_{i=6}^{N} \sum_{j<i} P_{ij}^{(l)} (i-j). \quad (24)$$

Figure 11 shows the 350M HOFA outlier pathway tracking standard MHA across all 24 depth layers (averaging 257.1 vs. 255.9 tokens, near the $N/4 = 256$ uniform-causal baseline at $N = 1024$). Conversely, the GLA pathway operates at a sharp local average of 3.9 tokens. Because d_l^{inlier} uses magnitude-normalized pseudo-probabilities rather than Softmax weights, these two figures provide a qualitative indicator of pathway specialization rather than an identical-metric comparison.

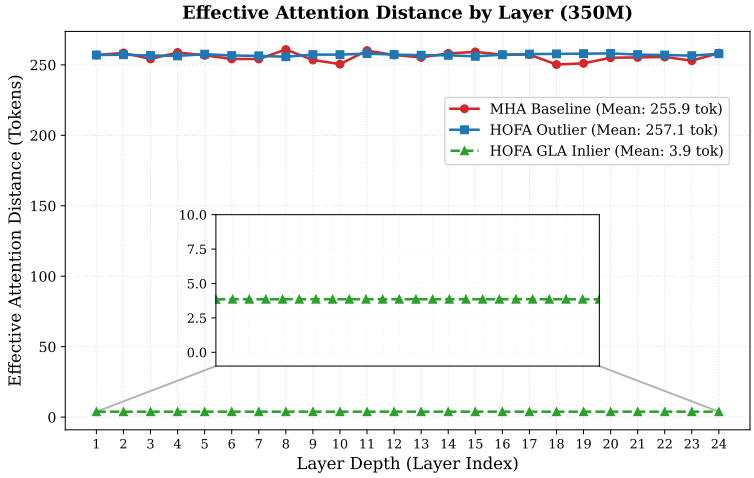
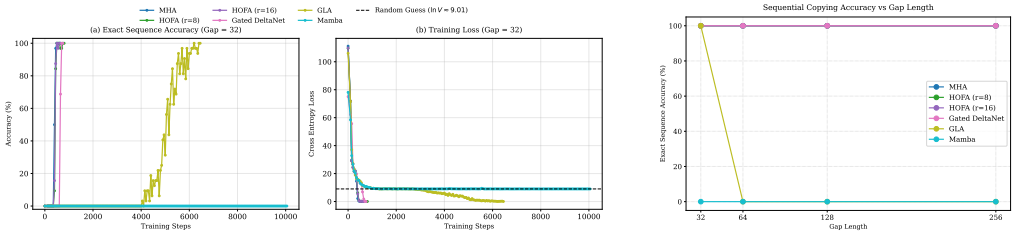


Figure 11: **Layer-Wise Effective Attention Distance (350M Model).** Comparison of standard MHA ($d_h = 128$), HOFA Outlier pathway, and HOFA GLA Inlier pathway across all 24 Transformer layers. The outlier pathway tracks MHA's layer-wise reach profile (257.1 vs. 255.9 tokens), while the GLA state absorbs local contextual background (3.9 tokens).

H Extended Multi-Token Block Copying Benchmarks

To isolate gap driven temporal state decay from pattern capacity limits, we evaluate 4-layer models ($L = 4, d_{\text{model}} = 128, H = 4, d_h = 32, \text{vocab} = 8192$) on multi-token block copying across noise gaps $G \in \{32, 64, 128, 256\}$ at pattern length $P = 2$. As shown in Figure 12a, Class I (MHA, HOFA) and Class II (Gated DeltaNet) converge within ~ 500 steps at $G = 32$, whereas Class III (GLA) requires ~ 5000 steps. Across longer gaps $G \geq 64$ (Figure 12b), GLA collapses to 0% accuracy from recurrent state collisions, while HOFA reliably maintains exact recall up to $G = 256$.



(a) Convergence ($G = 32$). (b) Accuracy vs. noise gap G.

Figure 12: **Multi-Token Block Copying Benchmarks** ($P = 2$)**. (a)** Class I (MHA, HOFA) and Class II (Gated DeltaNet) converge in ~ 500 steps at $G = 32$, while Class III (GLA) requires ~ 5000 steps and Class IV (Mamba-2) flatlines. **(b)** Exact recall accuracy vs. noise gap G; by $G \geq 64$, GLA collapses to 0%, while HOFA preserves retrieval up to $G = 256$.

I Orthogonal Architectural Extensions: Composite GQA and MLA Formulations

Multi-Head Latent Attention (MLA) [DeepSeek-AI, 2024] and Grouped-Query Attention (GQA) [Ainslie et al., 2023] are prominent strategies for KV-cache memory reduction, reducing memory footprint via low-rank projection or spatial key sharing. These mechanisms operate along structural dimensions orthogonal to HOFA's intra-head channel decomposition. Below, we propose conceptual composite formulations showing how HOFA can be combined with both mechanisms.

Composite HOFA–GQA. Grouped-Query Attention (GQA) achieves spatial compression by forcing G Query heads to share single Key and Value heads. In contrast, HOFA provides temporal

compression by offloading non-outlier channels to a recurrent GLA state. In a composite HOFA–GQA layer, G Query heads share a single r-dimensional Softmax Key head alongside a shared $(d_h - r)$-dimensional GLA inlier state, combining spatial head-sharing with temporal channel compression. A practical consideration is that aggressive spatial head sharing ($G \geq 8$) may constrain key diversity across heads, requiring appropriate selection of r.

Composite HOFA–MLA. Multi-Head Latent Attention (MLA) provides low-rank latent compression by projecting key-value states into compact latent spaces. Specifically, MLA requires a decoupled positional key path because low-rank compression and rotary embeddings do not commute; since HOFA already confines RoPE exclusively to the r-dimensional outlier subspace, this decoupling is inherently satisfied without additional positional projection machinery. Low-rank latent compression can therefore be applied directly to the unpartitioned Value stream V alone by caching a single shared low-rank latent vector $c_t^V \in \mathbb{R}^{d_c}$ per token (where $d_c < Hd_h$), reconstructing per-head value states at read time. This reduces the per-token cache footprint from $H(r + d_h)$ to $Hr + d_c$, showing that intra-head channel factorization is structurally compatible with latent compression. Retaining uncompressed outlier keys ($K_{\mathcal{O}}$) separates retrieval precision from value-stream compression, while decompressing value vectors introduces additional reconstruction operations during decoding.

These composite formulations are conceptual blueprints left to future empirical and hardware evaluation.

J Implementation & Triton Kernel Engineering

Prefill: Causal Loop Peeling. To optimize the exact outlier pathway under FlashAttention execution logic, we implement Causal Loop Peeling [Dao, 2023] in a custom Triton kernel; complete kernel implementations and evaluation suites are provided in the linked repository. Below is a streamlined structural excerpt of the fused forward kernel (`_fwd_kernel`) illustrating the two-phase loop peeling execution (unmasked block-sparse loop below diagonal and masked diagonal block):

```
@triton.jit
def _fwd_kernel(
    Q, K, V, Out, mix_g_ptr, sm_scale,
    stride_qz, stride_qh, stride_mix_g_n, ....,
    Z, H, N_CTX, r,
    BLOCK_M: tl.constexpr, BLOCK_N: tl.constexpr
):
    start_m = tl.program_id(0)
    off_hz = tl.program_id(1)

    # Unmasked blocks below diagonal
    for start_n in range(0, start_m * BLOCK_M, BLOCK_N):
        k = tl.load(k_ptrs, mask=mask_qk[:, None], other=0.0)
        qk = tl.dot(q, k, allow_tf32=True) * sm_scale
        m_ij = tl.maximum(m_i, tl.max(qk, 1))
        p = tl.exp(qk - m_ij[:, None])
        l_i = l_i * tl.exp(m_i - m_ij) + tl.sum(p, 1)
        acc = acc * tl.exp(m_i - m_ij)[:, None] + tl.dot(p.to(v.dtype), v)
        m_i = m_ij

    # Masked diagonal block
    for start_n in range(start_m * BLOCK_M, (start_m + 1) * BLOCK_M, BLOCK_N
        ):
        k = tl.load(k_ptrs, mask=mask_k, other=0.0)
        qk = tl.dot(q, k, allow_tf32=True) * sm_scale
        qk = tl.where(offs_m[:, None] >= offs_n_curr[None, :], qk, float("-
            inf"))
        m_ij = tl.maximum(m_i, tl.max(qk, 1))
        p = tl.exp(qk - m_ij[:, None])
        acc = acc * tl.exp(m_i - m_ij)[:, None] + tl.dot(p.to(v.dtype), v)
```

Fused Decoding. During generation ($N = 1$), we parallelize outlier attention across the sequence dimension using Split-K Flash-Decoding [Dao et al., 2023]. The kernel then combines the partitioned outlier attention while computing the inlier recurrent update, normalization, and inter-stream gating in a single pass.